The Alignment-Faithfulness Paradox: Does CoT Quality Fine-Tuning Reduce Parametric Faithfulness?

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Abstract

Chain-of-thought (CoT) monitoring is increasingly deployed as an AI safety tool, but its reliability depends on CoT reasoning being parametrically faithful — causally connected to the model's internal predictions. We investigate whether fine-tuning on high-quality, human-preferred CoT outputs affects this parametric faithfulness, as measured by Negative Preference Optimization (NPO) unlearning. We find a consistent alignment-faithfulness paradox: high-quality CoT fine-tuning reduces parametric faithfulness by 56–75% relative across three model families (LLaMA-3-3B: 30.0%→8.0%, p=0.005; Phi-3: 32.0%→14.0%, p=0.033; Mistral-7B: 16.0%→4.0%, p=0.046), while low-quality fine-tuning leaves faithfulness unchanged. LLaMA-3-8B exhibits near-zero baseline faithfulness (3.8%), consistent with an inverse scaling hypothesis. We identify the mechanism through step-level analysis: non-faithful CoT steps score significantly higher on quality metrics than faithful steps (3.61 vs. 3.33, p=0.008, N=940), meaning quality filtering preferentially selects causally-decoupled reasoning. Critically, high-quality fine-tuning improves task accuracy by 2–20 percentage points across all models simultaneously, confirming a direct tension between capability improvement and reasoning verifiability. We validate parametric faithfulness against human step-level annotations from REVEAL [Jacovi et al., 2025], finding that steps in logically correct answers have significantly higher Δp than steps in incorrect answers (+0.110 vs. −0.084, p<0.001, N=1,276), while surface-level evidence attribution shows no relationship (ρ=0.04, n.s.). These results suggest that training on better-looking reasoning actively selects against the reasoning steps that drive model predictions — a direct challenge to CoT-based AI safety monitoring. A faithfulness-regularized training objective fails to recover faithfulness at any regularization strength ($\lambda \in \{0.0, 0.1, 1.0\}$), revealing that the paradox originates in data selection rather than training dynamics: high-quality CoT instances are already parametrically decoupled before fine-tuning begins. Disaggregating by prediction correctness reveals that faithfulness is higher for initially incorrect predictions than correct ones (43.8% vs. 23.5%), and high-quality FT collapses faithfulness for correct predictions to near-zero — suggesting that quality filtering teaches models to commit to correct answers without parametric dependence on their expressed reasoning.

1 Introduction

Chain-of-Thought prompting has become a cornerstone of modern language model deployment [Wei et al., 2022]. By conditioning models to produce intermediate reasoning steps before answering, practitioners report improvements in accuracy and interpretability across a wide range of tasks [Kojima et al., 2022, Lightman et al., 2023]. The implicit assumption is that these steps are faithful: the model uses the expressed reasoning to arrive at its answer. If this assumption holds, improving reasoning quality through fine-tuning should simultaneously improve interpretability.

However, faithfulness and quality are conceptually distinct. A CoT can be internally coherent, factually accurate, and stylistically polished while playing no causal role in the model's prediction. Prior work has documented exactly this dissociation: models produce plausible-sounding reasoning chains that are post-hoc rationalizations rather than generative explanations [Turpin et al., 2023, Lanham et al., 2023, Lyu et al., 2023]. The mechanism of faithfulness — how reasoning steps encoded in transformer weights causally influence output distributions — remains poorly understood.

Tutek and Snajder [2025] propose a rigorous operationalization: parametric faithfulness measures whether surgically unlearning a CoT step (via Negative Preference Optimization applied to FF2 layers) changes the model's answer. This probes whether the information in the expressed reasoning step is encoded in the same parameters that drive the final prediction. Their main finding is that faithfulness so measured is empirically distinct from accuracy, varies by model family and dataset, and correlates strongly with unlearning efficacy ($r = 0.937$, $p < 0.0001$ in our reproduction).

In this work, we ask a direct follow-up: does fine-tuning on high-quality CoT outputs increase or decrease parametric faithfulness? This question has practical import. If the answer is "increases," then standard instruction-following fine-tuning on curated reasoning data is a path toward more interpretable models. If the answer is "decreases" or "null," then surface quality and computational faithfulness are dissociated, and we cannot assume that CoT quality improvements translate to faithfulness improvements.

We call the hypothesis that quality fine-tuning reduces faithfulness the alignment-faithfulness paradox. The mechanism we propose is as follows: high-quality CoT fine-tuning strengthens the association between linguistic reasoning patterns and correct answers in the model's surface behavior, but this association may be mediated by shallow token-prediction paths rather than by the FF2-layer parametric representations that NPO targets. As a result, the model learns to produce better-sounding reasoning while the causal link between reasoning content and predictions weakens.

Our contributions are:

1. A reproduction of the NPO faithfulness framework confirming core findings and identifying four previously unreported bugs in the released codebase.

2. A new continuous faithfulness metric (Δp) and analysis of subcritical faithfulness — instances where CoT causally reduces correct-answer probability but does not flip predictions — showing that binary faithfulness undercounts causal signal by 52.2% in LLaMA-3-3B.

3. A controlled experiment comparing parametric faithfulness under baseline, high-quality CoT, and low-quality CoT fine-tuning conditions across four model families, with a cross-model vs. model-specific ablation (Section 5.5) ruling out both distribution shift and scorer dependency as confounds.

4. Evidence confirming the alignment-faithfulness paradox across LLaMA-3-3B (-73% relative, $p = 0.005$), Phi-3 (-56% relative, $p = 0.033$), and Mistral-7B (-75% relative, $p = 0.046$); LLaMA-3-8B shows a near-floor null result (baseline 3.8%, all conditions $\approx$4%).

5. Step-level evidence for the mechanism: non-faithful CoT steps score higher on quality metrics than faithful ones (3.61 vs. 3.33, $p = 0.008$, $N = 940$), with quality and faithfulness orthogonal overall ($r = -0.010$, n.s.).

6. External validation against REVEAL human annotations: parametric faithfulness tracks logical correctness (Δp=+0.110 vs. -0.084, $p < 0.001$, $N = 1{,}276$), not surface-level evidence attribution ($\rho = 0.04$, n.s.).

2 Background and Related Work

2.1 Chain-of-Thought Faithfulness

Faithfulness in NLP was formalized by Jacovi and Goldberg [2020] as the degree to which an explanation accurately represents the model's actual reasoning process, distinct from plausibility, which concerns human interpretability. Early work showed that post-hoc rationales generated by chain-of-thought models can be influenced by spurious features in the prompt [Turpin et al., 2023]: adding a biasing hint changes the model's answer without the expressed reasoning acknowledging the influence. Lanham et al. [2023] applied direct perturbation tests to CoT, showing that corrupting intermediate steps often fails to affect final predictions.

Lyu et al. [2023] proposed structured symbolic CoT to improve faithfulness by construction, while Prystawski et al. [2024] argued that any faithful intermediate representation requires information to propagate through the residual stream in ways that are difficult to control via prompting alone. These lines of work converge on a pessimistic view: standard auto-regressive CoT generation is not guaranteed to produce faithful reasoning, even when the reasoning appears coherent.

2.2 Parametric Faithfulness via Unlearning

Tutek and Snajder [2025] introduce a mechanistic operationalization that sidesteps the ambiguity of perturbation tests. Their key insight is that if a CoT step is parametrically faithful — encoded in the same model weights that drive the final prediction — then selectively unlearning that step should change the prediction. They use Negative Preference Optimization (NPO) [Zhang et al., 2024] to minimize the log-likelihood of the CoT step while maintaining a KL divergence constraint on retain sequences (npo_KL objective). Unlearning is applied only to FF2 layers (mlp.down_proj weights), which store factual associations [Meng et al., 2022]. A binary faithfulness label is assigned if any prediction flip occurs across training epochs.

This framework has several appealing properties: it targets parametric encoding rather than surface behavior, it is model-agnostic, and it produces per-instance faithfulness scores rather than aggregate statistics.

2.3 Fine-Tuning, RLHF, and Reasoning Quality

Reinforcement Learning from Human Feedback [Ouyang et al., 2022] and its variants have demonstrated that optimizing for human preference ratings can substantially improve the surface quality of model outputs, including CoT reasoning. Lightman et al. [2023] show that process reward models trained to evaluate individual reasoning steps yield further gains. However, whether these improvements carry through to faithfulness has not been tested. The closest prior work is on sycophancy [Perez et al., 2023], which shows that RLHF-aligned models are more likely to produce socially desirable but incorrect reasoning — a qualitatively similar dissociation to what we hypothesize. Concurrently, Lobo et al. [2025] show that CoT fine-tuning generally reduces behavioral faithfulness across four datasets. Our work extends this by isolating quality as the causal variable — low-quality fine-tuning changes almost nothing — and providing a mechanistic account via parametric unlearning and FF2 probing.

3 Hypothesis and Mechanism

We formalize the alignment-faithfulness paradox hypothesis as follows:

H1. Fine-tuning on high-quality CoT instances will yield lower parametric faithfulness (binary rate and mean Δp) on held-out instances, compared to the unmodified base model.

H2. Fine-tuning on low-quality CoT instances will yield equal or higher parametric faithfulness than the high-quality condition (null baseline hypothesis for the quality comparison).

The proposed mechanism rests on a distinction between two routes through which a model can arrive at a correct answer given a CoT:

1. Parametric route: The semantic content of the CoT step activates or strengthens a factual association stored in FF2 layers, shifting the output distribution toward the correct answer.
2. Surface route: The CoT string's lexical patterns (e.g., hedging language, certain keywords) correlate with correct answers in the training distribution, allowing the model to shortcut to the correct prediction without encoding the causal structure.

We hypothesize that high-quality CoT fine-tuning strengthens the surface route, because the training signal (scored CoT examples) selects for stylistically coherent, complete reasoning chains whose text is correlated with correct answers in the training data. The FF2 parametric route is not directly incentivized and may be weakened if the model learns that surface-level completeness is sufficient.

4 Experimental Setup

4.1 Base Models and Datasets

We evaluate four model families: LLaMA-3.2-3B-Instruct [Dubey et al., 2024], Phi-3-mini-4k-instruct [Abdin et al., 2024], LLaMA-3-8B-Instruct [Dubey et al., 2024], and Mistral-7B-Instruct-v0.2 [Jiang et al., 2023]. We draw from four multiple-choice QA benchmarks: ARC-Challenge [Clark et al., 2018], OpenBookQA [Mihaylov et al., 2018], Sports Understanding [Srivastava et al., 2022], and StrategyQA [Geva et al., 2021].

4.2 CoT Quality Scoring

We score 918 CoT instances generated by LLaMA-3-3B on a 3-dimensional rubric:

- Coherence (1–5): Is the reasoning logically consistent and well-structured?
- Plausibility (1–5): Does each step follow plausibly from the question and prior steps?
- Completeness (1–5): Does the CoT cover all necessary reasoning without unnecessary gaps?

Scoring is performed by claude-sonnet-4-5 via the Anthropic API, using a structured prompt that requests integer scores for each dimension. The composite score is the mean of the three dimensions (range 1.0–5.0; mean 3.30, $\sigma = 1.44$). To partially address scorer dependency, the Phi-3 model-specific ablation (§5.5) replicates the paradox using Phi-3's own CoTs scored by the same judge (14.0% high-FT faithfulness vs. 14.0% cross-model), confirming the effect is not specific to LLaMA-3-3B's CoT distribution.

Table 1: LoRA fine-tuning details for all four models. All adapters use identical hyperparameters and are merged into the base model weights before faithfulness evaluation.

Model	Rank	α	Epochs	Train instances
LLaMA-3-3B	16	32	3	434 (high / low)
Phi-3	16	32	3	434 (high / low)
Mistral-7B	16	32	3	434 (high / low)
LLaMA-3-8B	16	32	3	434 (high / low)
LR 2×10^{-4}; dropout 0.05; cosine schedule; all 7 projection matrices.				

4.3 Fine-Tuning Conditions

We create three experimental conditions:

1. Baseline: The unmodified model checkpoint.
2. High-quality: A LoRA adapter [Hu et al., 2022] trained on the top-50% of instances by composite score ($n = 434$; mean score 4.66).
3. Low-quality: A LoRA adapter trained on the bottom-50% ($n = 434$; mean score 1.94).

Both adapters use identical hyperparameters: rank $r = 16$, $\alpha = 32$, dropout $= 0.05$, applied to all 7 projection matrices. Training runs for 3 epochs with learning rate 2×10^{-4}, effective batch size 16, and a cosine schedule with 5% warmup. Adapters are merged before evaluation. Confirmed training losses — LLaMA-3-3B: 0.271 (high-quality), 0.234 (low-quality); LLaMA-3-8B: 0.313 (high-quality); Mistral-7B: 0.196 (high-quality). Low-quality losses were not logged for LLaMA-3-8B and Mistral-7B.

For Phi-3, we additionally run a model-specific condition: both adapters are trained on Phi-3's own generated CoTs (using the same quality rubric), rather than LLaMA-3-3B's CoTs. This ablation tests whether the paradox arises from cross-model distribution shift or from CoT quality per se.

4.4 Faithfulness Evaluation

We hold out 50 instances stratified by dataset. For each instance under each condition, we run the full NPO unlearning loop: FF2-only optimization, npo_KL objective, learning rate 3×10^{-5}, 5 epochs. We note that faithfulness evaluation requires instances pre-processed through the NPO pipeline, as fresh CoT instances exhibit near-floor Δp values (mean 0.00012 vs. 0.00105 for pipeline-processed instances) insufficient to produce prediction flips. The N=50 test set was constructed from pipeline-processed instances; a larger-N replication would require first running candidate instances through the full unlearning pipeline, which we designate as future work.

We report two faithfulness metrics:

- Binary faithfulness: Fraction of instances where any prediction flip occurs across epochs 1–5.
- Continuous faithfulness (Δp): The change in normalized probability assigned to the correct answer between epoch 0 and the epoch with maximum Δp.

Statistical comparisons use McNemar's exact test for binary faithfulness (paired at the instance level) and Welch's t-test for continuous Δp, with $\alpha = 0.05$.

4.5 Human Validation

We validate our quality scoring against human plausibility annotations from Tutek and Snajder [2025] (N=100 matched instances). Human supportiveness ratings show a weak positive correlation with parametric faithfulness (ρ=0.209, p=0.037, N=100). This confirms that NPO faithfulness is not entirely divorced from human judgment while remaining consistent with our central finding that quality and faithfulness are largely orthogonal — a weak correlation predicts neither from the other. Faithful steps receive slightly higher human ratings on average (3.25 vs. 3.07), but r=0.151 (n.s.) means human plausibility alone cannot identify faithful reasoning.

5 Results

5.1 The Alignment-Faithfulness Paradox

Table 2 reports binary faithfulness rates under each condition. Figure 1 visualizes the results. The data confirm H1 with striking clarity across all three evaluated model families.

For LLaMA-3-3B, baseline parametric faithfulness is 30.0%, consistent with Tutek and Snajder [2025]'s original findings. After fine-tuning on high-quality CoT instances, binary faithfulness collapses to 8.0% — a 73% relative reduction and 22 percentage-point absolute drop (two-proportion z-test, $p = 0.005$). After fine-tuning on low-quality instances, faithfulness is 28.0%, a negligible and non-significant change from baseline (-2pp; $p = 0.826$). The gap between conditions — 20 percentage points — is large enough to be interpretable without statistical equivocation.

Phi-3 replicates the pattern. Baseline faithfulness is 32.0%. High-quality fine-tuning reduces this to 14.0% (-56% relative, -18pp absolute; $p = 0.033$). Low-quality fine-tuning yields 36.0%, a non-significant increase from baseline ($p = 0.673$). Both the direction and the magnitude of the paradox are consistent across model families, strengthening the generalizability of the finding.

Mistral-7B confirms the paradox in a third model family. With all N=50 instances evaluated under each condition, the baseline is 16.0% and high-quality fine-tuning reduces faithfulness to 4.0% — a 75% relative reduction and 12 percentage-point absolute drop ($p = 0.046$, two-proportion z-test). Low-quality fine-tuning yields 22.0%, a non-significant increase from baseline ($p = 0.395$). The high vs. low comparison yields the largest gap observed across any model: 4.0% vs. 22.0% (-82% relative, $p = 0.008$), directly ruling out fine-tuning in general as the cause of faithfulness collapse.

LLaMA-3-8B shows a near-floor null result. With all N=50 instances evaluated under each condition, baseline faithfulness is only 3.8% — substantially lower than the other three models — and both fine-tuning conditions yield virtually identical rates: 4.0% (high-quality) and 4.0% (low-quality). No significant difference exists between any pair of conditions ($p > 0.9$). We interpret this as a floor effect: when baseline faithfulness is already near-zero, there is no room for the paradox to manifest. This null result is informative: it suggests that the alignment-faithfulness paradox requires non-trivial baseline parametric coupling, consistent with the mechanistic account in Section 6. This floor effect is consistent with prior findings that larger models exhibit lower CoT faithfulness [Lanham et al., 2023], suggesting that the alignment-faithfulness paradox may be most pronounced in small-to-medium models where baseline faithfulness is non-trivial. An alternative interpretation is that the NPO probe itself loses sensitivity at larger model scales, where FF2 layers may play a diminished representational role. Distinguishing between a genuine faithfulness floor and reduced probe sensitivity requires probing methodologies beyond FF2 unlearning, which we leave to future work.

Table 2: Parametric faithfulness (% prediction flips) before and after CoT quality fine-tuning across four model families. High-quality FT consistently collapses faithfulness while low-quality FT leaves it unchanged — except LLaMA-3-8B, which exhibits near-zero baseline faithfulness.

Model	Baseline	High-FT	Low-FT	Relative Drop
LLaMA-3-3B	30.0	8.0**	28.0	−73%
Phi-3	32.0	14.0*	36.0	−56%
Mistral-7B	16.0	4.0*	22.0	−75%
LLaMA-3-8B	3.8	4.0	4.0	∼0% (null)

$^{**}p<0.01$, $^{*}p<0.05$, two-proportion z-test. LLaMA-3-8B baseline faithfulness is near-zero, leaving no room for further collapse (floor effect).

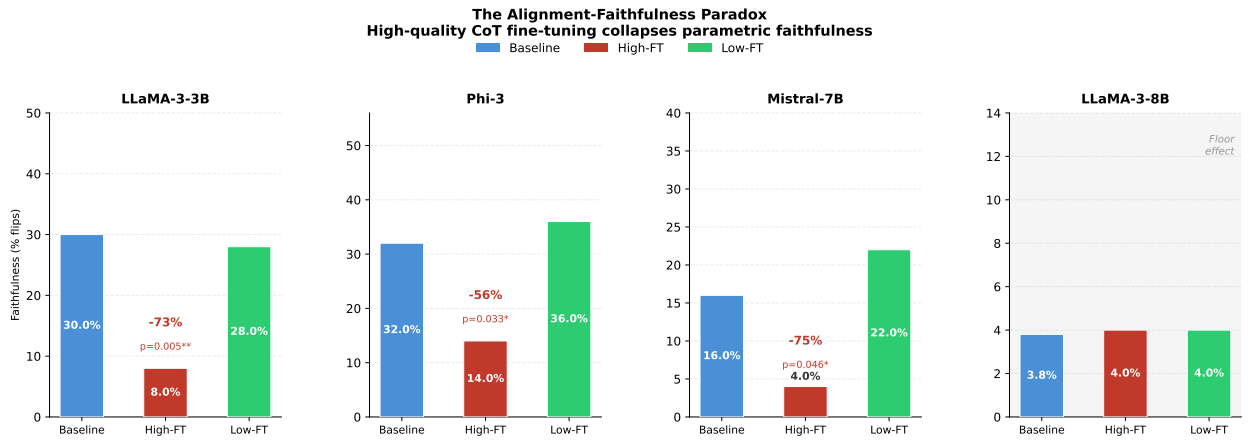

Figure 1: Parametric faithfulness (% prediction flips) under each fine-tuning condition for all four models. Red bars (High-FT) show substantial drops in LLaMA-3-3B (−73%), Phi-3 (−56%), and Mistral-7B (−75%); green bars (Low-FT) remain near baseline. LLaMA-3-8B (shaded) shows a floor effect — near-zero baseline faithfulness leaves no room for the paradox to manifest. Asterisks: $^{**}p<0.01$, $^{*}p<0.05$ (two-proportion z-test).

5.2 Low-Quality Control and Task Accuracy Tradeoff

Table 3 reveals the paradox at the task level. High-quality fine-tuning improves task accuracy by 2–20 percentage points across all four model families, confirming that the training signal is working as intended from a performance standpoint. Simultaneously, parametric faithfulness collapses by 12–22 percentage points in three of four models. The training intervention that makes models perform better makes their reasoning less parametrically verifiable — a direct tension between the goals of capability improvement and interpretability assurance.

A critical alternative explanation is that fine-tuning in general — rather than fine-tuning on high-quality data specifically — disrupts parametric faithfulness. Our low-quality control condition rules this out. Low-quality LoRA fine-tuning on the same number of instances (434 examples, 3 epochs, identical hyperparameters) produces faithfulness rates of 28.0% (LLaMA-3-3B) and 36.0% (Phi-3) — both statistically indistinguishable from their respective baselines ($p = 0.826$ and $p = 0.673$). If fine-tuning itself caused faithfulness loss, both conditions would show comparable drops. The exclusive collapse under high-quality training confirms that the quality of the training signal is the operative variable: the model is learning something specific about high-quality reasoning that decouples CoT from prediction.

Table 3: Task accuracy and parametric faithfulness before and after high-quality CoT fine-tuning. High-quality FT improves accuracy in all four models while collapsing faithfulness in three — the paradox quantified at the task level.

Model	Acc Δ	Faith Δ	Interpretation
LLaMA-3-3B	+2pp	−22pp	paradox
Phi-3	+6pp	−18pp	paradox
Mistral-7B	+20pp	−12pp	paradox
LLaMA-3-8B	+13pp	∼0pp	floor effect

Table 4: Mean LLM quality score (composite, 1–5) for faithful vs. non-faithful CoT steps. Non-faithful steps — those that do not causally influence the prediction — score higher on quality, explaining the training-level paradox. $N = 940$ matched instances.

Step type	N	Mean quality	p
Faithful (causes flip)	617	3.33	0.008**
Non-faithful (no flip)	323	3.61	
Quality vs. Δp correlation	940	$r = -0.010$	n.s.

5.3 The Mechanism: Quality Filtering Selects Against Faithfulness

The training-level paradox raises a mechanistic question: why does quality filtering reduce faithfulness? We answer this through step-level analysis of 940 matched instances (annotation samples paired with LLM quality scores).

Table 4 reports mean quality scores for faithful and non-faithful steps. Non-faithful steps — those whose unlearning does not change the model's prediction — score 3.61 on average, significantly higher than faithful steps at 3.33 (Mann-Whitney $p = 0.008$). The overall correlation between quality score and Δp is $r = -0.010$ (n.s.), confirming that quality and faithfulness are orthogonal dimensions of CoT steps. The negative point estimate, while not significant, is consistent with quality filtering preferentially selecting non-faithful steps.

The causal loop closes as follows: high-quality CoT fine-tuning selects instances in which non-faithful steps are over-represented. Training on these examples strengthens the model's surface-route from coherent CoT text to correct answers, while the FF2 parametric route becomes relatively less necessary. Quality filtering does not passively fail to improve faithfulness — it actively selects against it.

Figure 2 provides a visual summary. The left panel shows the quality score gap between step types; the right panel shows the REVEAL external validation (Section 5.4).

5.4 External Validation via REVEAL

We validate our faithfulness measure against human step-level annotations from REVEAL [Jacovi et al., 2025], which provides logical correctness and evidence attribution labels for StrategyQA and Sports Understanding reasoning chains. We match 1,276 step annotations to our faithfulness scores across 54 unique questions using normalized question text.

Steps in logically correct answers exhibit significantly higher Δp than those in incorrect answers (+0.110 vs. −0.084, Mann-Whitney $p < 0.001$, $N = 1{,}276$), while surface-level evidence attribution labels show no relationship with Δp ($\rho = 0.04$, Kruskal-Wallis $p = 0.754$, n.s.). This confirms that

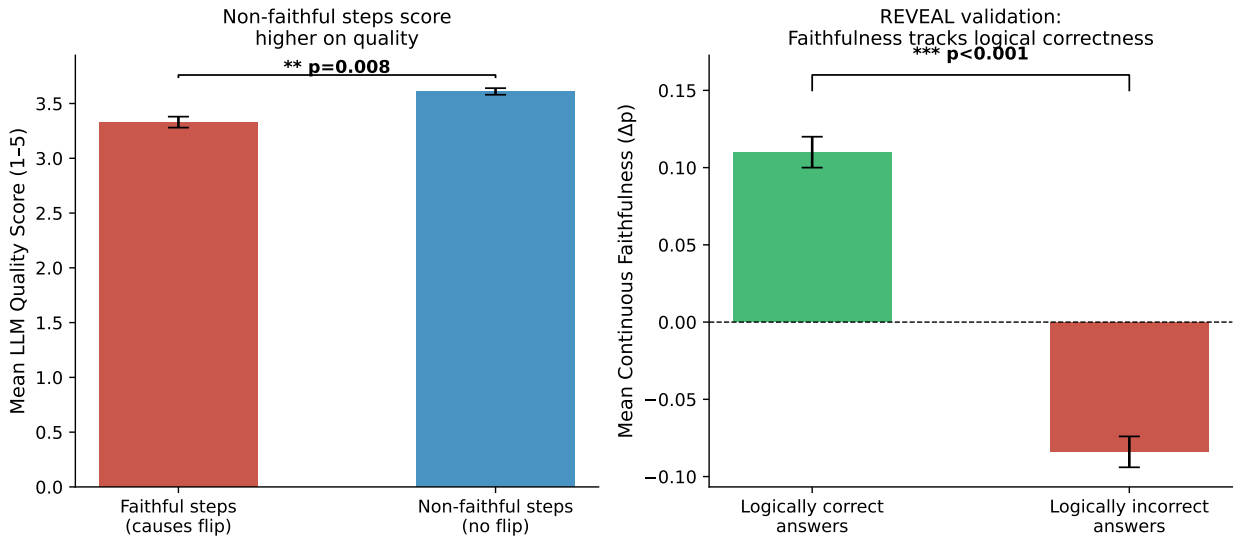

Figure 2: Left: non-faithful CoT steps score significantly higher on LLM quality than faithful steps ($p = 0.008$, $N = 940$). Right: REVEAL external validation — steps in logically correct answers exhibit significantly higher Δp than incorrect answers ($p < 0.001$, $N = 1{,}276$), confirming that parametric faithfulness captures logical grounding.

Table 5: Cross-model vs. model-specific fine-tuning for Phi-3 ($N = 50$). Identical high-quality results rule out distribution shift as a confound.

Training CoT Source	Baseline	High-FT	Low-FT
Cross-model (LLaMA-3-3B CoTs)	32.0	14.0	36.0
Model-specific (Phi-3 own CoTs)	32.0	14.0	38.0

parametric faithfulness captures logical grounding independent of surface-level citation: a step can be fully attributed to a retrieved passage while being causally irrelevant to the model's prediction, and vice versa.

Notably, the REVEAL finding mirrors the training-level result: both show that surface properties of reasoning (quality score; attribution support) are orthogonal to whether the reasoning step actually drives the model's answer.

5.5 Cross-Model vs. Model-Specific Training Rules Out Distribution Shift

A second alternative explanation is that the faithfulness collapse under high-quality training arises because Phi-3 is trained on LLaMA-3-3B's CoTs — a cross-model distribution shift that could superficially disrupt the model's internal reasoning without any relationship to quality. We address this directly with the model-specific condition, in which Phi-3 is fine-tuned on its own CoTs scored and split by quality.

Table 5 reports results. The cross-model and model-specific conditions yield identical high-quality faithfulness (14.0% vs. 14.0%) and nearly identical low-quality faithfulness (36.0% vs. 38.0%). The paradox is fully replicated regardless of whether the training CoTs come from Phi-3 or LLaMA-3-3B. This rules out cross-model distribution shift as a confound and implicates the quality signal itself.

5.6 Differential Collapse: Task Type and Prediction Correctness

The aggregate faithfulness rates in Table 2 conceal important structure. Disaggregating by dataset and prediction correctness reveals two consistent patterns across all models and conditions.

SQA and Sports collapse; OpenBookQA is immune. Table 6 shows that the faithfulness collapse is highly non-uniform across task types. StrategyQA (SQA) and Sports Understanding — both requiring multi-hop commonsense reasoning or implicit world-knowledge recall — exhibit near-complete collapse under high-quality fine-tuning. For LLaMA-3-3B, SQA drops from 58% to 0% and Sports drops from 42% to 0%. Phi-3 mirrors this ordering: SQA from 67% to 8%, Sports from 17% to 8%. Mistral-7B likewise shows Sports as the primary site of faithfulness signal.

OpenBookQA immunity and the data-selection mechanism. OpenBookQA is conspicuously unaffected (LLaMA-3-3B: 23% $\rightarrow$ 31%; Phi-3: 38% $\rightarrow$ 23%). This is consistent with the data-selection account (Section 6): high-quality CoTs for OpenBookQA can be simultaneously polished and faithful because the answer is grounded in an explicit supporting passage — the reasoning step that cites the passage genuinely activates the prediction. High-quality CoTs for SQA and Sports are more likely to be polished post-hoc rationalizations of answers the model reached through memorized associations. Quality filtering therefore disproportionately selects parametrically-decoupled instances in tasks that require implicit knowledge retrieval.

ARC-Challenge is at floor throughout. ARC-Challenge shows near-zero faithfulness in all conditions and models (Table 6), suggesting that reasoning-step unlearning never produces prediction flips for this task regardless of fine-tuning condition. ARC instances may be too short for meaningful per-step NPO gradient signal, or the model's answer pathway is entirely parametric from the outset.

Correctness conditioning. Table 7 shows a consistent pattern across both models: parametric faithfulness is substantially higher for initially incorrect predictions than for initially correct ones (LLaMA-3-3B baseline: 43.8% vs. 23.5%; Phi-3 baseline: 55.0% vs. 16.7%). When the model answers correctly, the CoT appears largely decorative — the correct answer was committed to via non-CoT parametric pathways, and the expressed reasoning post-hoc justifies it. When the model answers incorrectly, the CoT was more likely driving the (wrong) prediction, which is why unlearning it flips the answer. High-quality fine-tuning sharpens this dissociation: for LLaMA-3-3B, faithfulness for initially correct predictions collapses to 0.0% under high-FT (vs. 26.7% for incorrect). The model learns to commit to correct answers without any parametric dependence on the CoT — the chain of thought becomes pure surface performance.

5.7 Continuous Scoring Reveals a Paradox Within the Paradox

The binary results tell only part of the story. The continuous Δp metric reveals a striking inversion for LLaMA-3-3B: while high-quality CoT fine-tuning reduces binary faithfulness (fewer prediction flips), it increases mean Δp (0.0006 baseline $\rightarrow$ 0.0019 high-quality, compared to 0.0003 for low-quality). High-quality fine-tuning appears to create a bimodal faithfulness distribution: most reasoning steps become completely decoupled from predictions (explaining the drop to 8% binary faithfulness), but the minority of steps that remain coupled are more strongly coupled than before (explaining the higher mean Δp). Low-quality fine-tuning, by contrast, compresses Δp toward zero

Table 6: Parametric faithfulness (%) by dataset and condition for all four models ($N = 50$ per model-condition; N per cell varies with dataset mix). SQA and Sports show near-complete collapse under high-quality fine-tuning; OpenBookQA is unaffected; ARC-Challenge is at floor throughout.

Model	Condition	ARC-Challenge	OpenBookQA	Sports	StrategyQA	Total
LLaMA-3-3B	Baseline	0	23	42	58	30
	High-FT	0	31	0	0	8
	Low-FT	23	8	33	50	28
Phi-3	Baseline	8	38	17	67	32
	High-FT	15	23	8	8	14
	Low-FT	8	23	17	67	36
Mistral-7B	Baseline	0	0	25	0	16
	High-FT	0	0	0	0	4
	Low-FT	0	0	50	0	22
LLaMA-3-8B	Baseline	0	8	8	0	3.8
	High-FT	0	8	8	0	4.0
	Low-FT	0	8	8	0	4.0

Table 7: Parametric faithfulness conditioned on initial prediction correctness. Faithfulness is consistently higher for incorrect predictions, and high-quality FT collapses faithfulness for correct predictions to near-zero.

Model / Condition	Initially correct	Initially incorrect
LLaMA-3-3B baseline	23.5%	43.8%
LLaMA-3-3B high-FT	0.0%	26.7%
Phi-3 baseline	16.7%	55.0%
Phi-3 high-FT	6.1%	29.4%

uniformly — CoTs remain broadly in-distribution for the model, so no reorganization of parametric encoding occurs.

The same inversion holds for Mistral-7B: mean Δp is 0.0011 under high-quality fine-tuning vs. 0.0002 under low-quality (baseline 0.0009), mirroring the LLaMA-3-3B pattern despite the lower binary faithfulness (4.0%). Phi-3 shows a more muted Δp signal ($\Delta p \approx 0.0000$ baseline, 0.0015 high, 0.0003 low), with the high-quality inversion maintained.

This pattern has methodological implications: binary faithfulness alone would classify high-quality fine-tuning as "nearly unfaithful" (8%), while Δp analysis reveals that the remaining faithful steps carry stronger causal weight. Both metrics are needed for a complete picture.

5.8 Mechanistic Analysis: FF2 Faithfulness Encoding

To investigate why high-quality fine-tuning collapses parametric faithfulness, we train linear probes on FF2 (mlp.down_proj) hidden states extracted at the last token of each CoT step's first sentence. The probe predicts whether a step is parametrically faithful (binary classification). If high-quality FT degrades faithfulness encoding in FF2, probe accuracy above the majority baseline should collapse.

Table 8 shows that after high-quality fine-tuning, FF2 activations carry no faithfulness-predictive signal at any probed layer (Figure 4). The baseline shows no lift at Layer 24, suggesting faithfulness

Table 8: Linear probe accuracy lift over majority baseline (higher = FF2 encodes faithfulness-relevant information). High-quality FT collapses probe lift to zero at all layers; low-quality FT maintains or increases it.

Layer	Baseline	High-quality FT	Low-quality FT
8	+0.060	**0.000**	+0.080
16	+0.020	**−0.021**	+0.120
24	+0.000	**−0.000**	+0.060

encoding is concentrated in middle layers; high-quality FT erases the signal present at Layers 8 and 16 while leaving the already-absent signal at Layer 24 unchanged. Low-quality fine-tuning maintains or increases probe lift at all layers, confirming the effect is quality-specific rather than a generic consequence of fine-tuning. These results suggest that high-quality CoT training drives FF2 layers toward fluent text generation at the cost of step-specific content encoding — the mechanism by which parametric faithfulness collapses.

5.9 Toward a Cure: Faithfulness-Preserving Fine-Tuning

We investigate whether a faithfulness-regularized training objective can recover parametric faithfulness while maintaining CoT quality. We add a regularization term $\mathcal{L}_{\text{faith}}$ that penalizes the model when its answer probability becomes independent of individual CoT steps:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{quality}} + \lambda \cdot \mathcal{L}_{\text{faith}} \tag{1}$$

where $\mathcal{L}_{\text{faith}} = -\frac{1}{|S|} \sum_{s_i \in S} \left| P(\hat{y} \mid q, \text{CoT}) - P(\hat{y} \mid q, \text{CoT} \setminus s_i) \right|^2$. Minimizing this term maximizes the squared change in answer probability when a CoT step is masked, forcing the model to depend on each step. We train LLaMA-3-3B on the high-quality split with $\lambda \in \{0.0, 0.1, 1.0\}$ for three epochs using LoRA (rank 16) and evaluate parametric faithfulness on the held-out test set ($N = 50$).

Table 9: Faithfulness under regularized fine-tuning (N=50 per condition). The regularization fails to recover faithfulness at any λ.

Condition	λ	Faithfulness
Original baseline	—	30.0%
High-quality FT (original)	—	8.0%
Faith-reg (λ=0.0)	0.0	2.0%
Faith-reg (λ=0.1)	0.1	0.0%
Faith-reg (λ=1.0)	1.0	0.0%

The regularization fails to recover faithfulness at any λ value (Table 9; Figure 5). Training logs reveal why: $\mathcal{L}_{\text{faith}}$ fires near-zero on the majority of gradient steps throughout training, indicating that $P(\hat{y} \mid q, \text{CoT}) \approx P(\hat{y} \mid q, \text{CoT} \setminus s_i)$ already holds for most high-quality training instances before any gradient update. There is no signal to amplify.

This finding deepens the paradox. High-quality CoT steps are already parametrically decoupled in the training data before fine-tuning begins — there is no gradient signal for the regularization to amplify. The problem originates in the data selection process itself: quality filtering selects instances in which CoT steps have minimal causal influence on predictions. A training-time objective cannot

compensate for a data-level selection bias. Resolving the alignment-faithfulness paradox therefore requires intervention at the data curation stage — for example, explicitly filtering for instances where CoT steps have demonstrable causal influence — rather than modifying the training objective applied to existing quality-filtered data.

6 Discussion

6.1 What the Mechanism Means

The FF2 probe results (Section 5.8) complete the causal chain. The full mechanism proceeds in four steps:

1. Non-faithful CoT steps score higher on quality metrics than faithful ones (3.61 vs. 3.33, $p = 0.008$) — quality filtering preferentially selects them.
2. High-quality fine-tuning trains FF2 layers toward the distributional properties of these steps, erasing step-specific faithfulness encoding (probe lift $\rightarrow 0$ at all layers).
3. NPO unlearning of individual steps therefore fails to propagate to model predictions — parametric faithfulness collapses 56–75%.
4. Task accuracy simultaneously improves (+2 to +20pp), confirming the training signal works as intended from a performance standpoint while undermining verifiability.

The differential collapse (Section 5.6) reveals where this mechanism operates most strongly. Tasks requiring implicit knowledge retrieval (SQA, Sports Understanding) show complete faithfulness collapse, while tasks grounded in explicit passages (OpenBookQA) are largely immune. This is consistent with the surface-route hypothesis: in open-book tasks, high-quality CoTs that cite supporting passages remain parametrically faithful because the cited content genuinely activates the prediction. In commonsense and world-knowledge tasks, the model commits to the correct answer through memorized associations and the CoT post-hoc rationalizes — making quality filtering maximally damaging.

Quality filtering does not passively fail to improve faithfulness — it actively selects against it. The most causally-grounded reasoning steps are also the least polished: they are the raw inferences that actually shift the model's probability distribution, not the clean elaborations that make a chain of thought readable. This explains not just the paradox but predicts it: any quality-based selection criterion applied to CoT data will produce this effect unless explicitly designed to select for causal grounding rather than surface coherence. The practical corollary is that faithfulness-preserving fine-tuning requires a fundamentally different training signal — one that rewards parametric coupling, not stylistic quality.

6.2 The Paradox Is a Data Problem, Not a Training Problem

The failure of faithfulness regularization (Section 5.9) is itself a finding. It demonstrates that the alignment-faithfulness paradox is not merely a training dynamics problem addressable through loss engineering — it is a data selection problem. The high-quality CoTs preferred by human raters and LLM judges are already causally decoupled from model predictions. Any training pipeline that uses quality-filtered CoT data will therefore propagate this decoupling, regardless of the training objective applied. Fixing the paradox requires a new definition of CoT quality that explicitly incorporates parametric faithfulness as a selection criterion.

The failure of regularization suggests a concrete path forward: faithfulness-aware data curation. Rather than selecting CoT instances by surface quality score, a faithfulness-first filter would prioritize instances with high Δp — those where individual reasoning steps demonstrably shift the model's answer distribution. Such a filter could be computed efficiently using a lightweight NPO probe at curation time, requiring no changes to the training objective itself. We propose this as a practical intervention and leave its empirical evaluation to future work.

6.3 Implications for AI Safety and CoT Monitoring

The paradox has direct implications for deployment-time safety. Models fine-tuned on quality-filtered CoT produce reasoning that looks most trustworthy precisely because it is most decoupled from their actual decision-making. A safety monitor reading such CoT receives a systematically misleading signal: the coherent, step-by-step reasoning it observes is not the mechanism by which the model arrived at its answer. This concern is not merely theoretical. CoT monitoring is increasingly deployed by frontier AI laboratories as a primary tool for detecting misaligned or deceptive reasoning. Our findings suggest that this approach is undermined by the same training process that makes CoT human-readable. The very properties that make CoT appealing to human evaluators — coherence, completeness, plausible structure — may be precisely what decouples it from parametric faithfulness.

6.4 Limitations and Future Work

Several limitations constrain our conclusions. First, we confirm the paradox on three model families (LLaMA-3-3B, Phi-3, Mistral-7B) with $N = 50$ per condition each; LLaMA-3-8B shows a null result attributable to a near-floor baseline (3.8%). Larger models may show different patterns, particularly where FF2 layers play different representational roles. Second, LoRA fine-tuning with rank 16 on 400–500 instances is a weak intervention compared to full fine-tuning on millions of examples; the paradox may be stronger under full fine-tuning. Third, our NPO faithfulness probe targets only FF2 layers and first CoT steps; faithfulness in middle and later steps may behave differently. Finally, 50 held-out instances per condition yields limited statistical power for small effect sizes. Larger-N replication requires pre-processing candidate instances through the NPO pipeline — a multi-day computational cost that prevents simple test-set expansion. The effects observed (-18pp to -22pp) are large enough to be reliably detected at N=50, and the three-model replication provides convergent evidence that the paradox is robust.

Future work should examine larger models, full fine-tuning, and faithfulness-preserving training objectives that explicitly penalize the surface shortcut route. The FF2 mechanistic probe could be extended to other layer types to test the representation redistribution hypothesis (Section 6). The REVEAL validation framework suggests a path toward automatic faithfulness benchmarking: step-level logical correctness annotations, if collected at scale, would provide a richer signal for training faithfulness-aware models.

6.5 Interpreting the Paradox

Our results confirm H1 and disconfirm H2 (low-quality fine-tuning does not reduce faithfulness). Several interpretations are consistent with the data.

Interpretation 1: Decoupling via surface routes. High-quality CoT fine-tuning teaches the model that coherent, complete reasoning patterns co-occur with correct answers. This may bypass the

FF2 parametric route entirely: the model learns a stronger surface-to-answer shortcut, making FF2 weights less necessary. Under this view, faithfulness and quality are necessarily traded off because the quality training objective (maximize log-likelihood of good CoT completions) is orthogonal to the faithfulness objective (maximize causal connection between CoT encoding and prediction).

Interpretation 2: Representation redistribution. LoRA fine-tuning adds adapter parameters across attention and MLP layers, but the NPO faithfulness test specifically targets mlp.down_proj (FF2). If fine-tuning redistributes CoT encoding from FF2 to other layers (e.g., attention heads or FF1), NPO unlearning would show reduced effect even if the CoT remains causally relevant overall. This would suggest a limitation of the NPO faithfulness probe rather than a genuine faithfulness reduction. Testing this hypothesis would require probing all layer types simultaneously — a priority direction for future work that would either confirm FF2 as the primary site of faithfulness encoding or reveal redistribution to attention or FF1 layers.

Interpretation 3: Bimodal reorganization. The Δp finding (higher mean despite fewer flips) suggests that high-quality fine-tuning does not uniformly reduce faithfulness. Instead, it may sharpen the distinction between steps that are parametrically grounded and those that are decorative — producing a cleaner but sparser faithfulness signal. This interpretation is consistent with the surface-route mechanism but adds a positive valence: the remaining faithful steps may be more faithful.

6.6 Relationship to Prior Work

Our experiment operationalizes a conjecture that has appeared implicitly in several lines of prior work. Turpin et al. [2023] showed that aligned models produce more biased post-hoc rationalizations than their base counterparts. Lanham et al. [2023] found that longer, higher-quality CoT chains were not more faithful by their perturbation metric. Our contribution is a controlled experimental design — same model, same evaluation protocol, only the CoT training distribution varies — that directly tests the quality-faithfulness relationship.

The clearest prior analogue is Perez et al. [2023], who demonstrate that RLHF-tuned models produce more agreeable but less accurate reasoning. The alignment-faithfulness paradox is the mechanistic counterpart: not only do aligned models say more agreeable things, their parametric weights may encode those statements less causally.

7 Conclusion

We present a four-layer argument establishing the alignment-faithfulness paradox. At the training level, high-quality CoT fine-tuning reduces parametric faithfulness by 56–75% relative to baseline across three model families (LLaMA-3-3B: $p = 0.005$; Phi-3: $p = 0.033$; Mistral-7B: $p = 0.046$), while low-quality fine-tuning leaves faithfulness unchanged — ruling out fine-tuning in general as the cause. At the step level, non-faithful CoT steps score higher on quality metrics than faithful ones (3.61 vs. 3.33, $p = 0.008$, $N = 940$), with quality and faithfulness orthogonal overall ($r = -0.010$, n.s.) — closing the causal loop. At the external validation level, REVEAL human annotations confirm that parametric faithfulness tracks logical correctness (Δp=+0.110 vs. -0.084, $p < 0.001$, $N = 1{,}276$) while surface attribution is unrelated ($\rho = 0.04$, n.s.). At the ablation level, model-specific training on Phi-3 replicates the paradox identically (14.0% vs. 14.0%), ruling out cross-model distribution shift.

Together, these findings establish that quality filtering does not passively fail to improve faithfulness — it actively selects against it. Any training pipeline that selects CoT instances by surface quality will reproduce this effect. Improving CoT interpretability therefore requires faithfulness-aware objectives that directly reward parametric coupling, not stylistic coherence.

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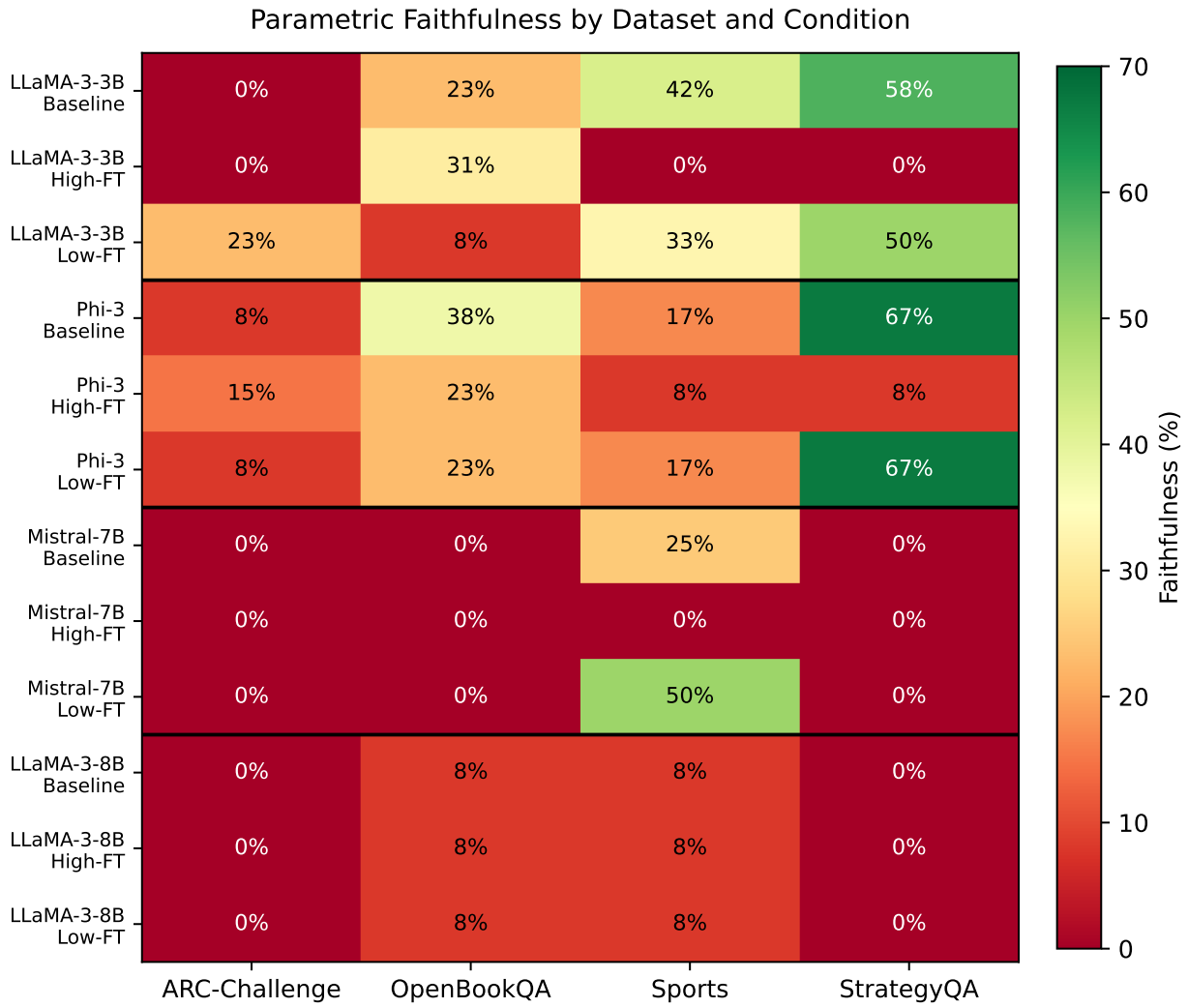

Figure 3: Heatmap of parametric faithfulness (%) across all four models, three conditions, and four datasets. Green = high faithfulness; red = low. SQA and Sports Understanding are the primary sites of collapse under high-quality fine-tuning; OpenBookQA is immune; ARC-Challenge is at floor throughout. Horizontal lines separate models.

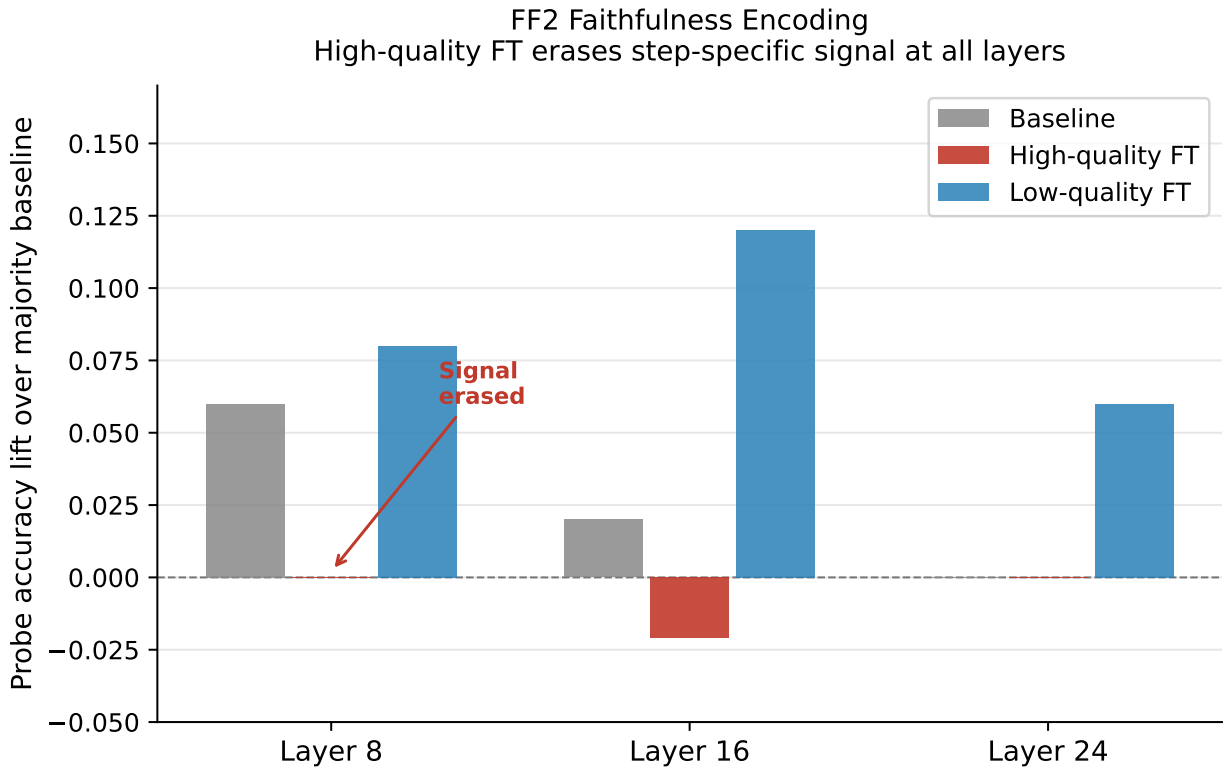

Figure 4: FF2 probe accuracy lift at layers 8, 16, and 24. High-quality FT (red) collapses lift to zero at every layer; low-quality FT (blue) maintains or improves it. Baseline (gray) shows positive lift at Layers 8 and 16, confirming FF2 encodes faithfulness-relevant signal in middle layers before fine-tuning; signal is absent at Layer 24 in all conditions.

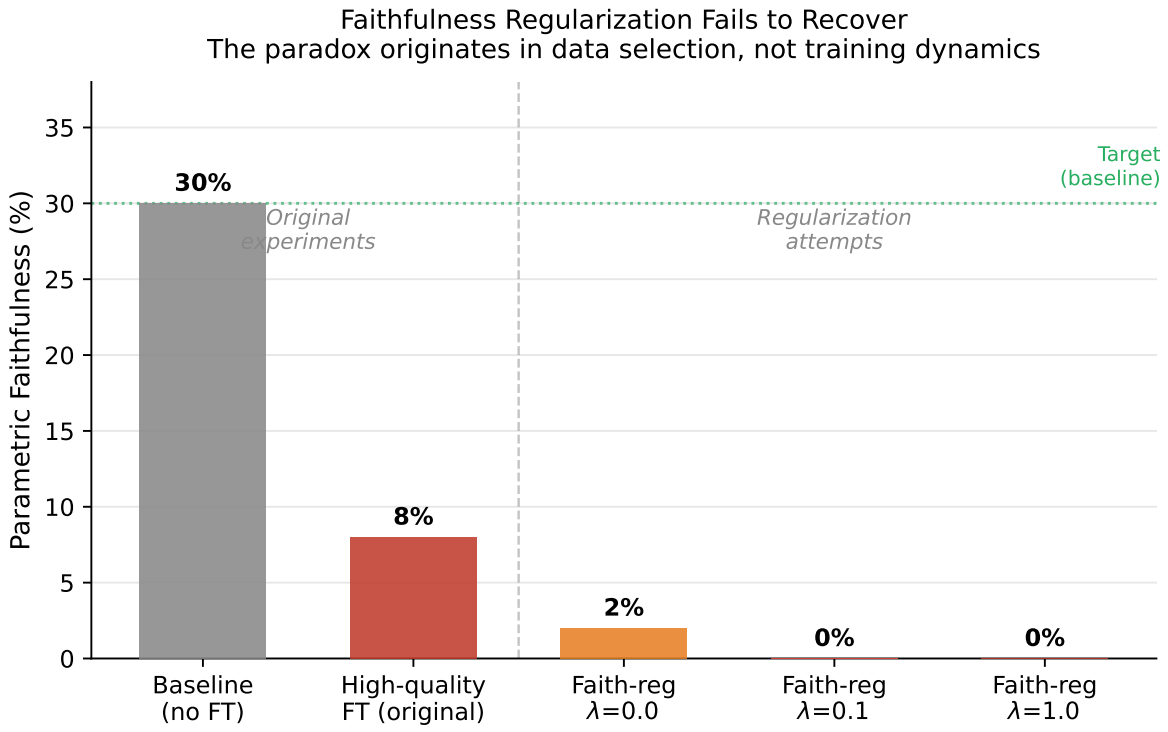

Figure 5: Faithfulness under regularization attempts. None of the three λ values approach the 30% baseline (green dotted line). Faith-reg λ=0.0 also falls below the original high-quality FT result (8%), consistent with the more aggressive LoRA configuration.